



US 20250287155A1

(19) **United States**

(12) **Patent Application Publication**
Scheek et al.

(10) **Pub. No.: US 2025/0287155 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **SPEAKER UNIT**

Publication Classification

(71) Applicant: **Mayht Holding B.V.**, Goleta, CA (US)

(72) Inventors: **Timothy Ruben Scheek**, Rotterdam (NL); **Naphur van Apeldoorn**, Amsterdam (NL)

(51) **Int. Cl.**

H04R 9/06 (2006.01)

H04R 1/28 (2006.01)

H04R 9/02 (2006.01)

(52) **U.S. Cl.**

CPC **H04R 9/063** (2013.01); **H04R 1/2819**

(2013.01); **H04R 9/022** (2013.01); **H04R**

2201/028 (2013.01); **H04R 2400/11** (2013.01)

(21) Appl. No.: **19/097,700**

(22) Filed: **Apr. 1, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/040,218, filed on Feb. 1, 2023, filed as application No. PCT/EP2021/071285 on Jul. 29, 2021, now Pat. No. 12,302,079.

(30) **Foreign Application Priority Data**

Aug. 3, 2020 (WO) PCT/EP2020/071790

Nov. 5, 2020 (EP) 20205819.4

(57)

ABSTRACT

A speaker unit with a speaker frame and two membranes arranged in the speaker frame. A first membrane radiates in a major acoustic radiation direction substantially perpendicular to a major plane of the speaker unit. A second membrane has a secondary acoustic radiation direction different from the major acoustic radiation direction. Two drive units are positioned within the speaker frame, attached to the two membranes, and positioned coaxial to each other at the same height in the speaker frame. An acoustic duct provides for a closed acoustic channel from the second membrane in the secondary acoustic radiation direction to a secondary surface of the speaker unit in a same plane as the major plane of the speaker unit.

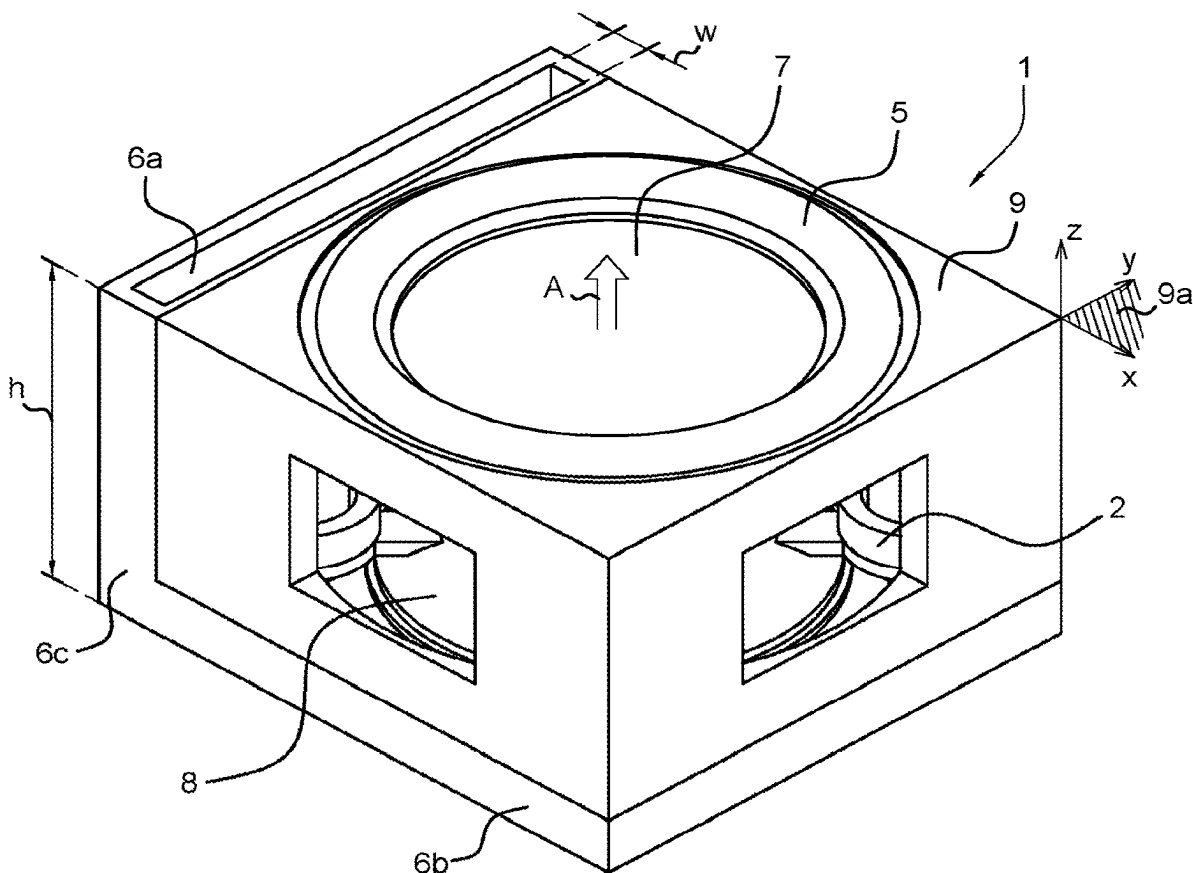


Fig. 1

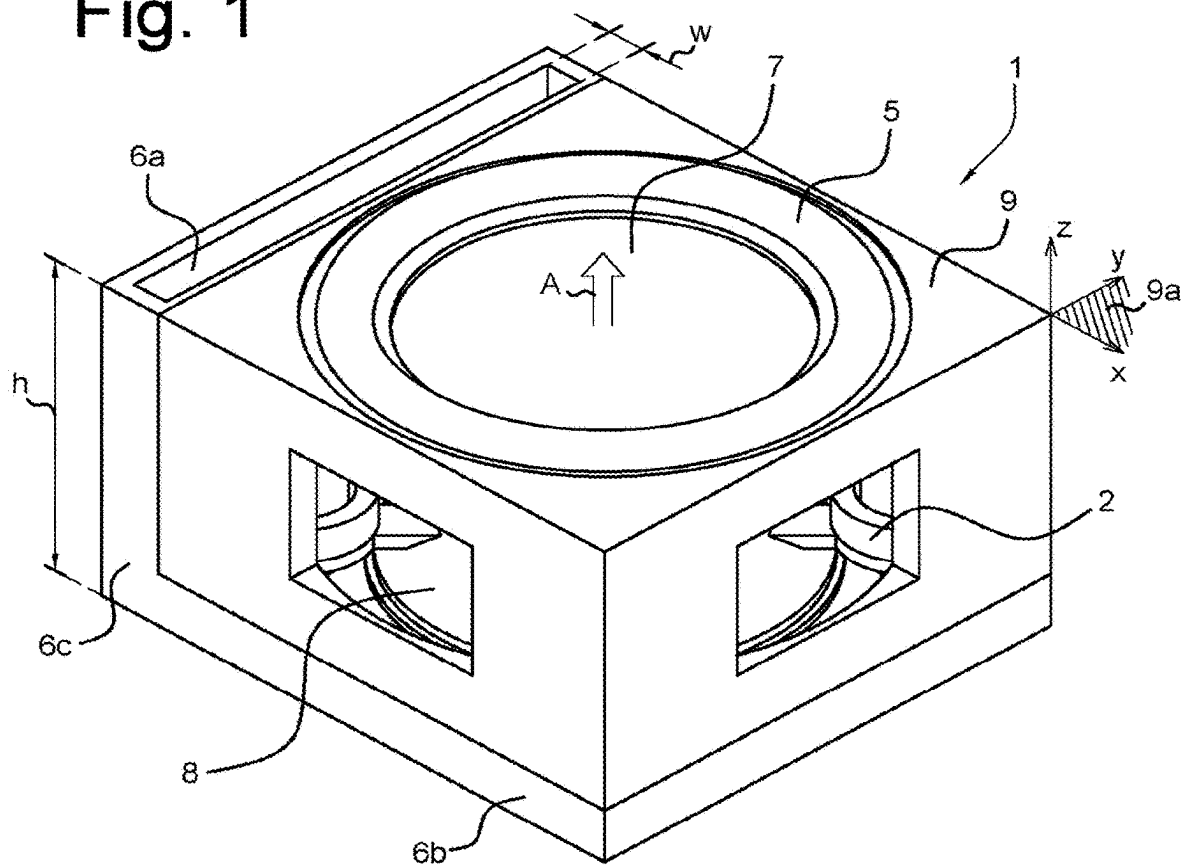


Fig. 2

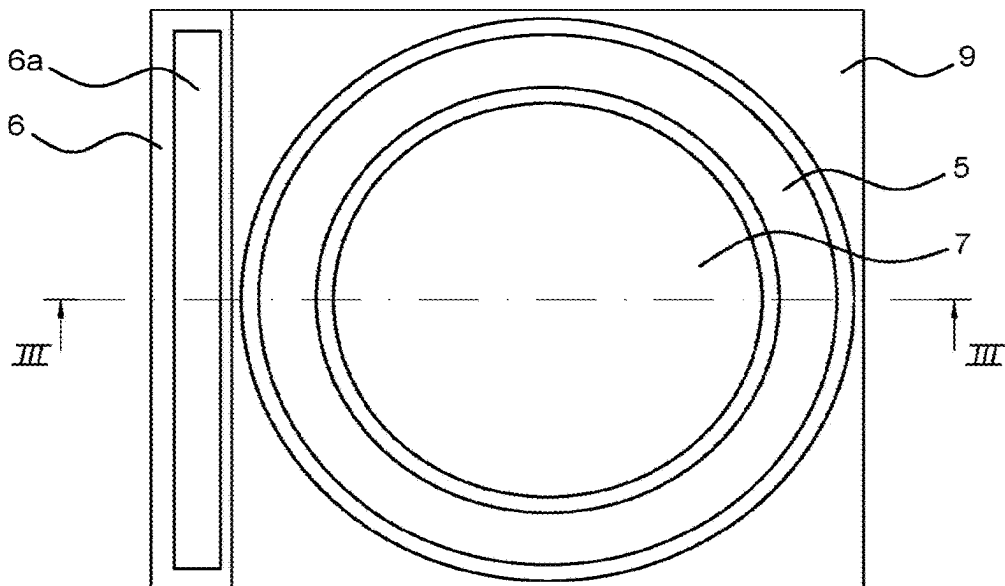


Fig. 3

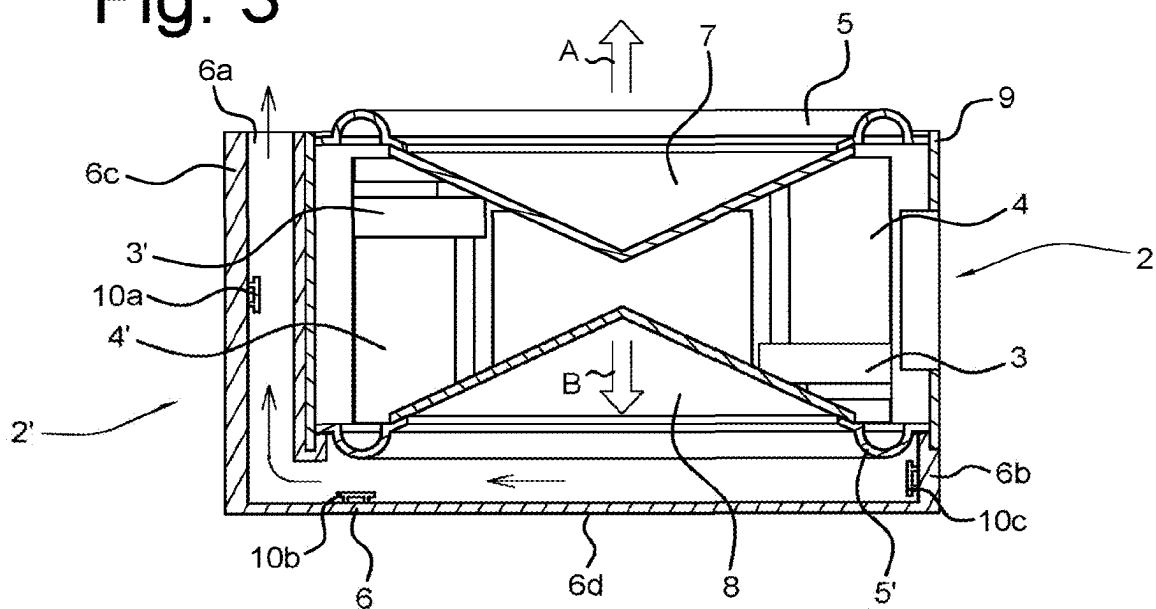


Fig. 4

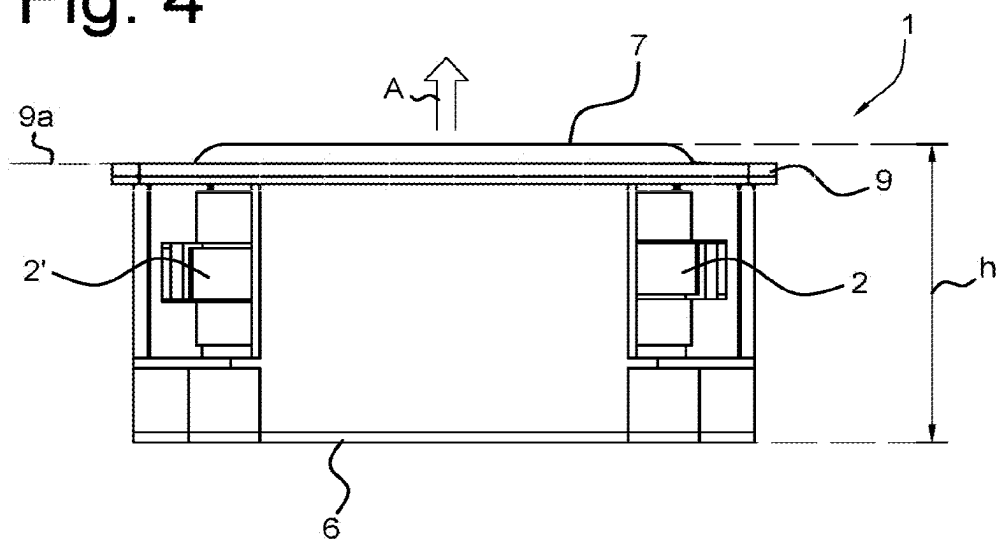


Fig. 5

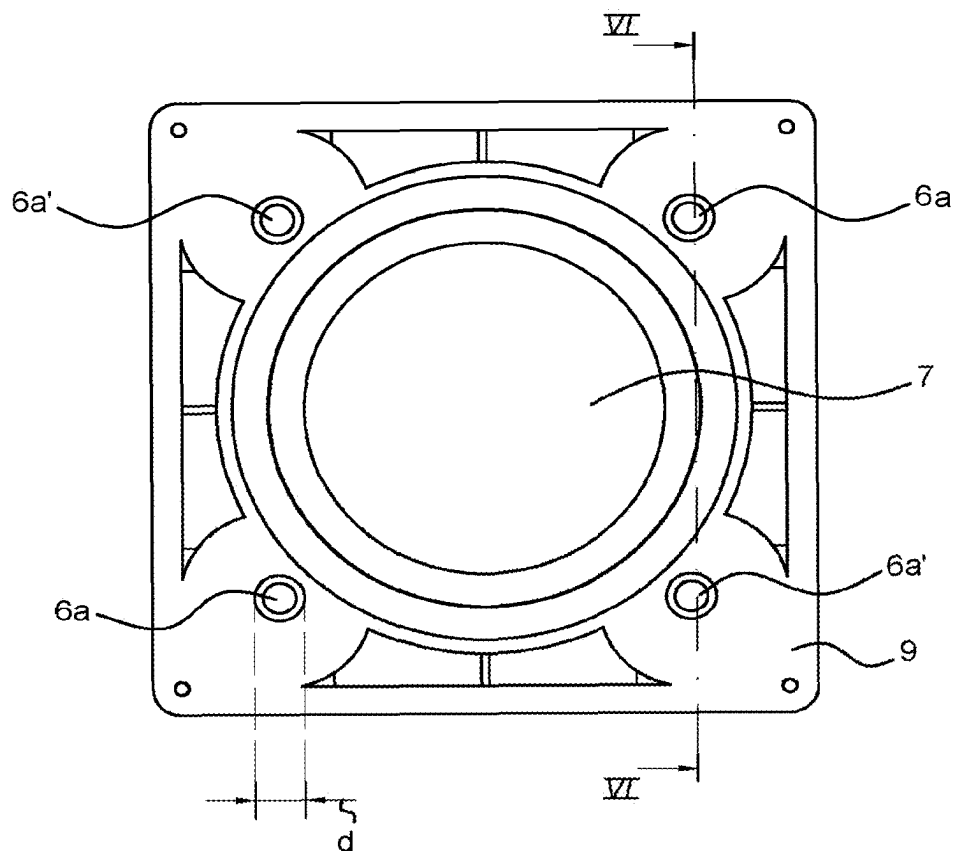
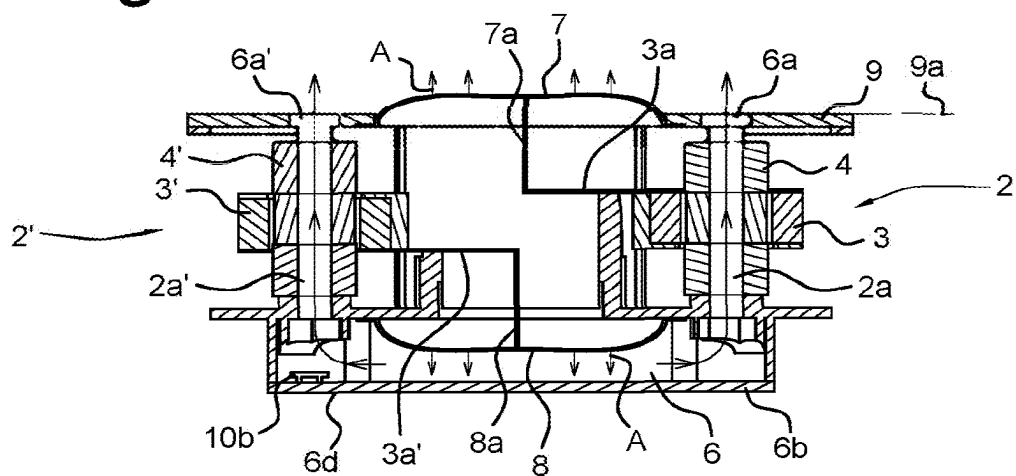


Fig. 6



SPEAKER UNIT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/040,218, filed Feb. 1, 2023, which is a 371 U.S. national phase application of International Application No. PCT/EP2021/071285, filed Jul. 29, 2021, which claims priority to International Application No. PCT/EP2020/071790, filed Aug. 3, 2020, and to European Patent Application No. 20205819.4, filed Nov. 5, 2020, each of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to a speaker unit comprising a speaker frame, two membranes arranged in the speaker frame, a first membrane of the two membranes having a major acoustic radiation direction substantially perpendicular to a major plane of the speaker unit, and a second membrane of the two membranes having a secondary acoustic radiation direction different from the major acoustic radiation direction, and two drive units positioned within the speaker frame and attached to the first and second membranes, respectively.

BACKGROUND ART

[0003] US patent publication US2010/0232637 discloses a speaker device having two opposed speakers positioned in a speaker box, wherein the two speakers are mechanically coupled and the speaker box is provided with an opening portion.

[0004] US patent publication US2007/0154044 discloses a loudspeaker system having multiple spherical enclosures, each housing a pair of (opposed) transducers.

[0005] US patent publication US2012/237077 discloses an opposing dual-vented woofer system. A vented speaker driver assembly is described utilizing a speaker driver having a pole piece that defines therein a vent that is not covered by a dust cap. The frame of the speaker driver is configured to be mounted to a structure's surface such that the pole piece of the driver is located within the structure's interior space. The vented speaker driver is configured to be utilized in a back-to-back vented driver assembly wherein the sound of more than one speaker driver is achieved in a footprint of only one driver and with minimized generation of physical vibrations from the assembly.

[0006] US patent publication U.S. Pat. No. 5,821,471 discloses speaker housings having various configurations, e.g. for directing backward aimed sound from a membrane in a housing towards a front side of the speaker.

[0007] International patent publication WO2019/086357 discloses a speaker unit having two opposing moving membranes.

[0008] International patent publication WO02/052892 discloses a speaker unit having a drive unit provided with air ducts.

SUMMARY OF THE INVENTION

[0009] The present invention seeks to provide a speaker unit having an improved performance in space efficiency, power, and freedom of air displacement directivity, as compared to existing speaker units. The speaker unit further

provides for improved cooling of electronic components of the speaker unit in a further group of embodiments.

[0010] According to the present invention, a speaker unit as defined above is provided, wherein the first and second membranes are arranged in opposite configuration in the speaker frame and the secondary acoustic radiation direction is opposite to the major acoustic radiation direction, wherein the first and second membranes are coaxially aligned along the major and opposite second acoustic radiation direction, and wherein the two drive units are positioned coaxial to each other at the same height in the speaker frame and laterally displaced from the membranes in a side-by-side arrangement therewith. The speaker unit further comprises an acoustic duct providing a closed acoustic channel from the second membrane in the secondary acoustic radiation direction to a secondary surface of the speaker unit, the secondary surface being located in a same plane as the major plane of the speaker unit.

[0011] The present invention embodiments have a structure and mutual element orientation allowing to provide a self-balancing, more space efficient speaker unit for dual membrane units which have air displacement direction restrictions (e.g. vehicle doors, in-ceiling speakers, TV, or any speaker with single-sided air displacement restrictions).

SHORT DESCRIPTION OF DRAWINGS

[0012] The present invention will be discussed in more detail below, with reference to the attached drawings, in which

[0013] FIG. 1 shows a perspective view of a speaker unit according to an embodiment of the present invention;

[0014] FIG. 2 shows a top view of the speaker unit embodiment shown in FIG. 1;

[0015] FIG. 3 shows a cross sectional view of the speaker unit embodiment shown in FIG. 2 along the lines III-III;

[0016] FIG. 4 shows a side view of a speaker unit according to a further embodiment of the present invention;

[0017] FIG. 5 shows a top view of the speaker unit embodiment shown in FIG. 4;

[0018] FIG. 6 shows a cross sectional view of the speaker unit embodiment shown in FIG. 5 along the lines VI-VI.

DESCRIPTION OF EMBODIMENTS

[0019] The present invention will be explained in detail with reference to some exemplary embodiments shown in the drawings, which are only intended to show embodiments of the invention and not to limit the scope. The scope of the invention is defined in the annexed claims and by its technical equivalents. A person skilled in the art will understand that features, components, elements, etc. explicitly used to explain the invention can be substituted by technical equivalents unless otherwise stated. Moreover, separate features of different embodiments can be combined, even if not explicitly shown in the drawings or explained in the specification, unless such combination is physically impossible. The present invention will be discussed in more detail below, with reference to some drawings. The examples and embodiments described herein serve to illustrate rather than to limit the invention. The person skilled in the art will be able to design alternative embodiments without departing from the scope of the claims. Reference signs placed in parentheses in the claims shall not be interpreted to limit the scope of the claims. Items described as separate entities in

the claims or the description may be implemented as a single or multiple hardware items combining the features of the items described.

[0020] It is to be understood that the invention is limited by the annexed claims and its technical equivalents only. In this document and in its claims, the verb “to comprise” and its conjugations are used in their non-limiting sense to mean that items following the word are included, without excluding items not specifically mentioned. In addition, reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one”.

[0021] In the speaker unit embodiments of the present invention, two examples of which are shown in the FIGS. 1-6 and are discussed below, the following major elements are present (with reference numerals as indicated, and synonym terms between brackets):

- [0022]** 1. speaker unit (speaker, loudspeaker, loudspeaker device)
- [0023]** 2. drive unit (driver, motor)
- [0024]** 3. voice coil
- [0025]** 4. magnet assembly (at least two magnets)
- [0026]** 5. membrane suspension (surround, surround roll, flexible edge)
- [0027]** 6. acoustic duct (membrane duct)
- [0028]** 7. first membrane
- [0029]** 8. second membrane
- [0030]** 9. speaker frame

[0031] In general, the present invention relates to a speaker unit 1 comprising a speaker frame 9, (at least) two membranes 7, 8, (at least) two drive units 2, and an acoustic duct 6. The two membranes 7, 8 are arranged in the speaker frame 9, a first membrane 7 of the two membranes having a major acoustic radiation direction A substantially perpendicular to a major plane 9a of the speaker unit 1, and a second membrane 8 of the two membranes having a secondary acoustic radiation direction B different to the major acoustic radiation direction A. Note that the first and second membrane 7, 8 thus move up and down in the indicated acoustic radiation direction A, B, but that the sound from the speaker has a direction away from the membranes 7, 8. The two drive units 2 are positioned within the speaker frame 9 and attached to the two membranes 7, 8, the two drive units 2 being positioned coaxial to each other at the same height in the speaker frame 9. An acoustic duct 6 is present and provides a closed acoustic channel from the second membrane 8 in the secondary acoustic radiation direction B to a secondary surface 6a of the speaker unit 1, the secondary surface 6a being located in a same plane as the major plane 9a of the speaker unit 1. The acoustic duct 6 is thus arranged to redirect acoustic waves from the second membrane 8 to radiate in the same plane as the first membrane 7.

[0032] In a more specific embodiment, the two membranes 7, 8 are arranged in opposite configuration in the speaker frame, and the secondary acoustic radiation direction B is opposite to the major acoustic radiation direction A (i.e. 180°).

[0033] FIG. 1-3 show a first exemplary embodiment of the present invention speaker unit 1. FIG. 1 shows a perspective view of the speaker unit 1, FIG. 2 shows a top view, and FIG. 3 shows a cross sectional view of the speaker unit embodiment along the lines III-III in FIG. 2. The first membrane 7

is flexibly connected to the speaker frame 9 (at a front surface of the speaker unit 1) using a membrane suspension 5 (or surround, surround roll, flexible edge) allowing up and down motion of the first membrane 7 in the direction of the major acoustic radiation direction A. The second membrane 8 is likewise flexibly connected using a similar membrane suspension 5', allowing movement of the second membrane 8 in the direction of the secondary acoustic radiation direction B.

[0034] The speaker frame 9 may be arranged as a combination of top, bottom and four side walls, optionally provided with apertures to reduce overall weight of the speaker unit 1, as shown in the views of FIGS. 1 and 3. Alternatively, the speaker frame 9 may be provided as a combination of scaffold like elements, with appropriate attachment means for the elements of the speaker unit 1. The major plane 9a is indicated in FIG. 1 as coinciding with a front surface of the speaker unit 1, i.e. in the x-y plane of the indicated three dimensional axes x, y, z.

[0035] In this embodiment, the secondary surface 6a is located adjacent to a front surface of the speaker frame 9. The acoustic energy emanating from the second membrane 8 is redirected to the secondary surface 6a, and thus provided adjacent to the acoustic energy emanating from the first membrane 7, both in the major acoustic radiation direction A.

[0036] In the cross sectional view of FIG. 3, a further embodiment is shown, wherein the acoustic duct 6 has a first part 6b arranged at a predetermined distance from the second membrane 8 along the secondary acoustic radiation direction B, and a second part 6c at one or more sides of the speaker frame 9. In the embodiment shown, a single second part 6c is shown, but alternatively, the second part 6c is present at two, three, or even all four sides of the speaker frame 9. These chimney like embodiments allow to maintain very limited dimensions of the speaker unit 1, yet providing a highly efficient, front radiating arrangement.

[0037] FIG. 4-6 show a second exemplary embodiment of the present invention speaker unit 1. FIG. 4 shows a side view, and FIG. 5 shows a top view of the speaker unit 1 shown in FIG. 4. FIG. 6 shows a cross sectional view of the speaker unit 1 shown in FIG. 5 along the lines VI-VI.

[0038] In a group of embodiments similar to the embodiment shown in FIGS. 4-6, the secondary surface 6a is co-located with a front surface of the speaker frame 9. The acoustic energy emanating from the second membrane 8 is redirected to a plurality of secondary surfaces 6a, 6a', and thus provided adjacent to the acoustic energy emanating from the first membrane 7, both in the major acoustic radiation direction A. In the embodiment shown in FIG. 5, four secondary surfaces 6a, 6a' are provided equally distributed around the first membrane 7, allowing to still use a limited front surface area of the speaker unit 1.

[0039] As shown most clearly in the cross sectional view of FIG. 6, in a further embodiment, one or more of the (at least) two drive units 2 have an inner bore 2a, and part of the acoustic duct 6 is formed by the inner bore 2a of the one or more of the two drive units 2. As is clear from the top view of FIG. 5 and the cross sectional view of FIG. 6, in this embodiment two diagonally positioned drive units 2 are present to drive the first membrane 7, providing bores 2a in communication with the secondary surfaces 6a. Furthermore, two diagonally positioned drive units 2' are present to drive the second membrane 8, providing bores 2a' in com-

munication with the secondary surfaces 6a'. The surface of second membrane 8 (directed in the secondary acoustic radiation direction B), first part 6b, bores 2a, 2a', and secondary apertures 6a, 6a' thus form the acoustic duct 6 in this exemplary embodiment.

[0040] In the exemplary embodiment shown in FIG. 6, the drive units 2, 2' each comprise a voice coil 3, 3' and magnet assembly 4, 4'. The voice coils 3, 3' are mechanically driving the first membrane 7 and second membrane 8, respectively, via mechanical linkages 3a, 7a; 3a', 8a. In this embodiment, the membranes 7, 8 are implemented as flat surface membranes, but it will be apparent that other types of membranes can be utilized (e.g. cone shaped membranes similar to the ones shown in the FIG. 1-3 embodiment) with appropriate mechanical linkage means to the voice coils 3, 3'.

[0041] In prior art speaker systems, the dual opposing driver principle in the classic sense is used, wherein drivers are placed in a back-to-back position. The benefit of this architecture is that the opposing drivers cancel out mechanical vibrations of the enclosure of the speaker unit. Because of this cancellation, the enclosure is affected significantly less by the movement of the drivers, even if the enclosure is relatively light, of low rigidity and or small in relation to the drivers. The downside in these earlier prior art systems employing back-to-back positioned drivers, is that the footprint is bound by at least two times the depth of the identical drivers.

[0042] Converging the drivers into coaxially positioned drive units 2 is an efficient manner for decreasing the minimum amount of volume needed in the speaker design, as for example described in the published patent application WO2019/086357 of the present applicant, which is incorporated herein by reference. A further development of loudspeaker devices possibly having a low profile are described in International patent application WO2019/117706 of the present applicant, which is also incorporated herein by reference. The drive units 2 as applied in the present invention embodiments may also be implemented as the units described in International patent application WO2018/056814, which is also incorporated herein by reference.

[0043] It is noted that currently applied damper and port solutions are not usable in the converged driver architecture described above. When the membranes 7 make an excursion inwards (towards each other), the variable distance between the membranes 7 would create a problematic situation for a static centred port as e.g. known from US patent publication U.S. Pat. No. 8,452,041.

[0044] It is noted that the acoustic duct 6 as used in the present invention embodiments has a functionality which is acoustically and mechanically different from a ported speaker enclosure, where the port is used as a way of enhancing the bass performance or providing a band pass filter. Also, the air displacement which is redirected, comes from the side of the membrane 8 movement which is acoustically in phase and mechanically out of phase with the side of the other membrane(s) 7 which is (are) moving in free air. The acoustic duct 6 purpose is to redirect the air displacement of the second membrane 8 with as little effect on the acoustic output of the redirected membrane's air displacement as possible.

[0045] The speaker unit 1 may comprise a vented frame element, e.g. as top plate of the speaker frame 9 as shown in FIG. 5. Such a vented frame element provides space for air displacement in free air space, in the outward excursion

direction A of the first membrane 7 which is closest to the vented frame element). The acoustic duct 6 or air guide is provided to redirect the air displacement from the second membrane 8 towards a side which is not equal to the outward (outward meaning excursion away from the speaker unit 1) excursion direction B of the second membrane 8. The acoustic duct 6 provides a fully airtight connection between the second membrane 8 outward excursion direction and free air. The acoustic duct 6 has a frontal opening (total aperture area of secondary surface(s) 6a) towards free air, allowing sufficient air to move through without significant acoustical effects (band pass or bass enhancement effects).

[0046] According to the present invention, various embodiments of a speaker unit 1 are provided, wherein each of a plurality of drive units 2 comprises at least one voice coil 3 and a magnet assembly 4 with at least two magnets. To direct all air displacement towards a single surface or several surfaces providing the major acoustic radiation direction A, the acoustically in phase air displacement of at least one of the two membranes 7, 8 is redirected using the acoustic duct 6 towards a surface 6a which is not equal to the acoustically in phase outward excursion direction of the membrane 8 of which the air displacement is being redirected.

[0047] The combination of speaker frame 9, two membranes 7, 8 and drive units 2, 2' of the present invention embodiments could be placed in a loudspeaker cabinet, having at least one acoustic duct 6 which enable both membranes 7, 8 to displace air to free air (i.e. the speaker unit 1 exterior), where air displacement of at least one membrane 8 is guided through the acoustic duct 6 towards a side of the speaker unit 1 which is not equal to the outward excursion direction of the redirected membrane 8.

[0048] In a further embodiment, the first membrane 7 and second membrane 8 are cone shaped (see the FIG. 1-3 embodiment). This will provide an efficient speaker unit 1 in terms of obtainable sound pressure level and further speaker characteristics. Additionally, a flat shaped protection shield in the form of a mesh or plain surface of light material may be added. In an alternative embodiment, the first membrane 7 and second membrane 8 are flat (see the FIG. 4-6 embodiment).

[0049] In order to obtain an even more efficient speaker unit 1, the acoustic duct 6 has an inner surface arranged to guide acoustic waves in a further embodiment. This may be obtained by e.g. using a suitable (plastic) material with a proper acoustic properties.

[0050] In a further embodiment, the speaker unit 1 has an inner space delimited by the speaker frame 9 and the two membranes 7, 8, the two drive units 2 being positioned in the inner space. The voice coil 3 and magnet assembly 4 of the drive units 2 are then isolated from environmental air, and from the acoustic waves generated by the membranes 7, 8.

[0051] In order to obtain an efficient speaker unit 1, wherein also the second membrane 8 contributes significantly to the sound produced by the speaker unit 1, a smallest cross section of the acoustic channel 6 is larger than 1 cm² in a further embodiment. Such a dimensional limit will provide a sufficiently low acoustical damping to obtain a high enough sound pressure level emanating from the secondary surfaces 6a.

[0052] Alternatively, or additionally, in a further embodiment, a smallest width of the acoustic channel 6 is selected to be larger than 5 mm. E.g. a width w of the secondary

surface 6a as indicated in the embodiment shown in FIG. 1, or a diameter d of the secondary surfaces 6a, 6a' in the embodiment of FIG. 5, may be selected to be larger than 5 mm. For acoustic waves, a smallest dimension of the acoustic duct is most relevant to obtain a sufficiently low acoustic damping.

[0053] With respect to the previous two embodiments, it is noted that in alternative embodiments, where the speaker unit 1 is a very small speaker unit, the smallest cross section and/or smallest width can even be smaller.

[0054] In even further embodiments, a frontal surface area the secondary surface 6a is at least 10% of a frontal surface area of the second membrane 8. This has the effect that a sufficient high part of the sound energy produced by the second membrane 8 is guided towards the front side of the speaker unit 1, and contributes to the total sound pressure level that can be generated by the speaker unit 1.

[0055] To enable the present invention speaker unit 1 to be applied in many applications, such as in vehicle doors, as in-ceiling speaker, in television apparatus, etc., a total height h of the speaker unit 1 (see embodiment shown in FIG. 1 and in FIG. 4) is substantially equal to or smaller than four times a maximum peak-to-peak excursion of each of the (at least) two membranes 7, 8. The specific structure of the present invention embodiments using the coaxially positioned drive units 2 (i.e. in a side by side arrangement, laterally displaced from the membranes 7, 8) allows to keep the thickness dimension of the speaker unit 1 this limited.

[0056] In further advantageous embodiments, the speaker unit 1 of the present invention also provides for improved cooling of internal or external components of the speaker unit 1, such as the (at least) two drive units 2, but also e.g. one or more electronic components for driving the two drive units 2. To obtain this effect, at least part of the acoustic duct 6 is made from thermally conductive material.

[0057] Referring to FIGS. 3 and 6 as mentioned hereinabove, the acoustic duct 6 has the first part 6b arranged at a predetermined distance from the second membrane 8 along the secondary acoustic radiation direction B, and the second part 6c at one or more sides of the speaker frame 9. A single second part 6c is shown, but the second part 6c may also represent two, three or even all four sides of the speaker frame 9.

[0058] To prevent excessive heat build-up within the speaker frame 9 when the speaker unit 1 is in operation, an embodiment is provided wherein the acoustic duct 6 (e.g. first part 6b and/or the second part 6c) comprise thermally conductive material. This embodiment allows heat being generated in the speaker frame 9 by e.g. the two drive units 2 to be dissipated through improved thermal conductivity of the first and second parts 6b, 6c of the acoustic duct 6. Alternatively or additionally, the acoustic duct 6 can now also be used for cooling externally mounted components (e.g. electronic components) by using the moving air inside the acoustic duct 6 during operation (even if actual volume is limited, there is still exchange of heat energy via the air moving in the acoustic duct 6. So in this embodiment the first and second parts 6b, 6c now act as thermal heat sinks for the speaker unit 1.

[0059] It is worth noting that, since the second part 6b may comprise one, two, three or all four sides of the speaker frame 9, that a desired level of heat dissipation can be achieved by choosing which of the one, two, three and/or four sides should comprise thermally conductive material.

[0060] Thermal performance of the speaker unit 1 may be improved by considering an embodiment wherein the acoustic duct 6 further has a third part 6d as shown in FIGS. 3 and 6, wherein the third part 6d is arranged opposite or in front of the second membrane 8 at a predetermined distance therefrom. This third part 6d may be viewed as a back side/wall of the speaker unit 1 extending substantially parallel to the second membrane 8 and where the back side is circumferentially connected to the first and second parts 6b, 6c. Then, in an advantageous embodiment, the third part 6d may also comprise thermally conductive material and to act as a thermal heat sink to further dissipate heat being generated by the speaker unit 1, or an external component mounted on the third part 6d.

[0061] By taking the above into account, a combined embodiment can be envisaged wherein the acoustic duct 6 further has the third part 6d arranged opposite the second membrane 8 at a predetermined distance therefrom, and wherein the first part 6b, the second part 6c and/or the third part 6d comprise thermally conductive material.

[0062] In an exemplary embodiment, the thermally conductive material exhibits a thermal conductivity of at least 100 W/m*K. So when the first part 6b, the second part 6c and/or the third part 6d comprise e.g. aluminium or copper, then the thermal conductivity of at least 100 W/m*K would be achieved.

[0063] In an embodiment, as shown in FIGS. 3 and 6, the speaker unit 1 may comprise one or more electronic components 10a, 10b, 10c for e.g. driving the two drive units 2. These one or more electronic components 10a, 10b, 10c may include a speaker amplifier, filter circuitry, a power supply and/or any other electronics needed for the speaker unit 1. The components 10a, 10b, 10c are shown as internally mounted components (i.e. within the acoustic duct 6), but in further embodiments the components 10a, 10b, 10c, are mounted to an outside surface of the acoustic duct 6.

[0064] Because the one or more electronic components 10a, 10b, 10c may generate heat, an embodiment is conceivable wherein the one or more electronic components 10a, 10b, 10c are mounted in thermal contact with the acoustic duct 6 (e.g. on the first part 6b, the second part 6c and/or the third part 6d each of which comprises thermally conductive material accordingly. This embodiment effectively allows the one or more electronic components 10a, 10b, 10c to use the acoustic duct 6 as a thermal heat sink, wherein heat from the one or more electronic components 10a, 10b, 10c is dissipated through the thermally conductive first part 6b, second part 6c and/or third part 6d.

[0065] It is worth noting that also air cooling of the one or more electronic components 10a, 10b, 10c may occur in the embodiment wherein the one or more electronic components 10a, 10b, 10c are mounted inside the acoustic duct 6 on the thermally conductive first part 6b, second part 6c and/or third part 6d. So when the speaker unit 1 is in use, movement of air through the acoustic duct 6 will cool the one or more electronic components 10a, 10b, 10c arranged therein. Note that this air cooling also occurs when none of the first part 6b, second part 6c and/or third part 6d are thermally conductive.

[0066] In even further embodiments, also a further part of the speaker unit 1 which is in thermal contact with the acoustic duct 6 can be made from thermally conductive

material, such as a front surface of the speaker frame 9, allowing external components that need cooling to be mounted thereon.

[0067] The present invention has been described above with reference to a number of exemplary embodiments as shown in the drawings. Modifications and alternative implementations of some parts or elements are possible, and are included in the scope of protection as defined in the appended claims.

1. A loudspeaker assembly, comprising:
 - a frame having a first side portion and a second side portion;
 - a first membrane and a second membrane;
 - a first suspension element flexibly connecting the first membrane to the first side portion of the frame;
 - a second suspension element flexibly connecting the second membrane to the second side portion of the frame;
 - a first motor operably coupled to the first membrane, wherein the first motor is configured to drive the first membrane outward in a first direction;
 - a second motor operably coupled to the second membrane, wherein the second motor is configured to drive the second membrane outward in a second direction different from the first direction, and wherein the first and second motors are at least partially laterally displaced from the first membrane and the second membrane with respect to the first and second directions, respectively;
 - an aperture adjacent to the first side portion of the frame; and
 - an airtight acoustic channel extending from the aperture toward the second membrane, wherein the airtight acoustic channel is arranged to direct acoustic energy emanating from the second membrane toward the aperture and to cause the acoustic energy to be emitted from the aperture in a third direction different from the second direction.
2. The loudspeaker assembly of claim 1, wherein the third direction is the same as the first direction.
3. The loudspeaker assembly of claim 1, wherein the airtight acoustic channel has a first portion arranged at a predetermined distance from the second membrane along the second direction and a second portion at one or more sides of the frame.
4. The loudspeaker assembly of claim 2, wherein the second direction is opposite the first direction.
5. The loudspeaker assembly of claim 1, wherein the first motor and the second motor are arranged to drive the first and second membranes such that air displaced by outward movement of the second membrane is acoustically in phase and mechanically out of phase with air displaced by outward movement of the first membrane.
6. The loudspeaker assembly of claim 1, wherein one or more of the first motor and the second motor define an inner bore, and part of the airtight acoustic channel is formed by the inner bore of the one or more of the first motor and the second motor.
7. The loudspeaker assembly of claim 1, wherein the first membrane and second membrane are aligned with each other with respect to a first axis.
8. The loudspeaker assembly of claim 7, wherein the wherein the first and second motors are laterally displaced

from the first membrane and the second membrane with respect to a second axis that is perpendicular to the first axis.

9. The loudspeaker assembly of claim 1, comprising a third motor operably coupled to the first membrane, wherein the third motor is configured to drive the first membrane outward in the first direction, and a fourth motor operably coupled to the second membrane, wherein the fourth motor is configured to drive the second membrane outward in the second direction.

10. The loudspeaker assembly of claim 1, wherein the loudspeaker assembly has an inner space delimited by the frame and, the first motor and the second motor being positioned in the inner space.

11. The loudspeaker assembly of claim 1, wherein at least part of the airtight acoustic channel is at least partially made from thermally conductive material.

12. The loudspeaker assembly of claim 11, further comprising one or more electronic components for driving the first motor and the second motor, and wherein the one or more electronic components are mounted in thermal contact with the airtight acoustic channel.

13. The loudspeaker assembly of claim 1, comprising a further airtight acoustic channel extending from the aperture toward the first membrane, wherein the further airtight acoustic channel is arranged to direct acoustic energy emanating from the first membrane toward the aperture and to cause the acoustic energy to be emitted from the aperture in a fourth direction different from the first direction.

14. The loudspeaker assembly of claim 13, wherein the fourth direction is the same as the third direction.

15. The loudspeaker assembly of claim 13, wherein the fourth direction and the third direction are each different to both the first direction and the second direction.

16. The loudspeaker assembly of claim 13, wherein at least part of the airtight acoustic channel and at least part of the further airtight acoustic channel form a common airtight acoustic channel.

17. A playback device comprising:

- a frame having a first side portion and a second side portion;
- a first membrane and a second membrane;
- a first suspension element flexibly connecting the first membrane to the first side portion of the frame;
- a second suspension element flexibly connecting the second membrane to the second side portion of the frame;
- a first motor operably coupled to the first membrane, wherein the first motor is configured to drive the first membrane outward in a first direction;
- a second motor operably coupled to the second membrane, wherein the second motor is configured to drive the second membrane outward in a second direction different from the first direction, and wherein the first and second motors are at least partially laterally displaced from the first membrane and the second membrane with respect to the first and second directions, respectively;
- an aperture adjacent to the first side portion of the frame; and
- an airtight acoustic channel extending from the aperture toward the second membrane, wherein the airtight acoustic channel is arranged to direct acoustic energy emanating from the second membrane toward the aper-

ture and to cause the acoustic energy to be emitted from the aperture in a third direction different from the second direction.

18. The playback device of claim **17**, wherein the third direction is the same as the first direction.

19. A loudspeaker assembly for use in a wall or ceiling, the loudspeaker assembly comprising:

- a frame having a first side portion and a second side portion;
- a first membrane and a second membrane;
- a first suspension element flexibly connecting the first membrane to the first side portion of the frame;
- a second suspension element flexibly connecting the second membrane to the second side portion of the frame;
- a first motor operably coupled to the first membrane, wherein the first motor is configured to drive the first membrane outward in a first direction;

- a second motor operably coupled to the second membrane, wherein the second motor is configured to drive the second membrane outward in a second direction different from the first direction, and wherein the first and second motors are at least partially laterally displaced from the first membrane and the second membrane with respect to the first and second directions, respectively;

an aperture adjacent to the first side portion of the frame; and

an airtight acoustic channel extending from the aperture toward the second membrane, wherein the airtight acoustic channel is arranged to direct acoustic energy emanating from the second membrane toward the aperture and to cause the acoustic energy to be emitted from the aperture in a third direction different from the second direction.

20. The loudspeaker assembly of claim **19**, wherein the third direction is the same as the first direction.

* * * * *